
Detecting Anomalies in Toronto Urban Traffic via Poisson GLM, GRU, and KL Divergence

Omkumar Miteshbhai Patel ^{*} ¹

Abstract

Urban traffic flows exhibit non-linear dependencies and extreme outliers driven by large-scale events. Standard monitoring approaches often fail to distinguish between predictable high-volume events and true anomalies. This paper compares a classical probabilistic baseline and a neural sequence model for hourly traffic prediction and anomaly detection in Toronto. A Poisson generalized linear model (GLM) provides interpretable effects of temporal and event features, while a gated recurrent unit (GRU) captures nonlinear temporal dependence from recent history. We score anomalies using Poisson Kullback-Leibler (KL) divergence between observed counts and model-implied distributions, and we evaluate detections against a multi-criteria ground truth constructed from extreme values, hour-conditioned outliers, rapid changes, and high-traffic flags. Results show large forecasting gains for the GRU (54.2% lower RMSE than the GLM), but higher anomaly-detection F1 for the GLM under the chosen ground-truth definition on 5 years of Toronto traffic data. A case study on Blue Jays game days supports the view that predictable extremes should not be treated as anomalies.

1. Introduction

Urban traffic exhibits strong periodic structure alongside intermittent disruptions. Daily commuting patterns and scheduled events generate predictable fluctuations, while incidents, construction, and irregular demand introduce deviations. Consequently, traffic anomalies cannot be defined solely by high volume; they must be assessed relative to what is statistically expected given time, context, and recent

history.

Most operational traffic monitoring systems rely on fixed thresholds or simple percentiles. These methods frequently flag predictable peaks—such as rush hours or event-related surges—as anomalies, producing excessive false alarms. A principled anomaly detection framework must therefore distinguish between predictable extremes and genuinely unexpected behavior.

1.1. Modeling Perspective

Traffic volumes are nonnegative count data, naturally suited to probabilistic modeling. Poisson generalized linear models (GLMs) provide an interpretable baseline by relating expected traffic to temporal and event-based covariates. However, GLMs assume conditional independence across observations and cannot capture short-term temporal dependence beyond engineered features.

Neural sequence models offer a complementary approach. Gated Recurrent Units (GRUs) explicitly model temporal dependence and can learn nonlinear dynamics from recent traffic history. While these models often achieve superior predictive accuracy, they are rarely framed within probabilistic anomaly detection or compared directly against interpretable baselines.

Recurrent neural networks such as LSTM and GRU models have been shown to perform well for short-horizon traffic forecasting due to their ability to model temporal dependencies.

1.2. Approach and Contributions

This project adopts a dual-model framework that combines the strengths of both paradigms. A Poisson GLM serves as an interpretable probabilistic baseline, while a GRU captures nonlinear temporal structure using recent observations. Rather than defining anomalies via raw prediction error, we compute anomaly scores using Kullback–Leibler (KL) divergence between Poisson distributions implied by observed counts and model predictions.

We evaluate anomalies in two ways:

- (i) divergence between observations and each model’s pre-

¹Lassonde School of Engineering, York University, Toronto, Ontario, Canada. Correspondence to: Omkumar Miteshbhai Patel <patelom@yorku.ca>.

dicted distribution

(ii) divergence between the GLM and GRU predicted distributions. The second perspective highlights hours where a simple linear baseline and a temporally aware model fundamentally disagree.

1.3. Evaluation and Key Insight

Evaluating traffic anomalies is challenging due to the absence of authoritative labels. We address this by constructing a multi-criteria ground truth using extreme values, hour-conditioned outliers, rapid temporal changes, and high-traffic indicators. Models are trained on data from 2020–2024 and evaluated on 2025 as a blind test year to prevent leakage.

The results show that the GRU substantially improves the accuracy of the forecast, while the GLM achieves a higher anomaly-detection F1 under the built ground truth. A case study on Toronto Blue Jays game days reveals that the GRU assigns lower anomaly scores to recurring event-driven surges, supporting the interpretation that predictable extremes should not be treated as anomalies.

2. Data and Feature Construction

2.1. Dataset and Target Variable

The dataset consists of hourly traffic count observations indexed by location, date, and hour. Each row represents one location–hour measurement with an observed traffic volume $y_t \in \mathbb{Z}_{\geq 0}$. The modeling target is the hourly count y_t (denoted as `y_true` in the evaluation outputs). The raw data includes location identifiers and coordinates, along with timestamp fields that support temporal feature construction.

2.2. Preprocessing Pipeline

The preprocessing pipeline enforces a consistent hourly indexing and produces a supervised learning table for both the GLM and GRU. The pipeline performs:

- **Temporal normalization:** timestamps are parsed into (date, hour) and aligned to an hourly grid per location.
- **Data cleaning:** incomplete rows and malformed timestamps are removed. The cleaned table preserves location identity and hour-level traffic counts.
- **Feature derivation:** binary calendar and event features are computed from the timestamp and external event schedule.

The evaluation table used downstream contains the fields visible in the exported evaluation outputs, including `location_name`, `date`, `hour`, `Match_Day`,

`is_weekend`, `is_peak`, `High_Traffic`, and `y_true`, with model predictions stored as `lambda_glm` and `lambda_gru`.

2.3. Feature Set

We use a compact feature set that supports two requirements: interpretability for the GLM and robust temporal context for the GRU.

- **Hour** (0–23) captures diurnal structure.
- **Weekend indicator** captures weekend regime shifts.
- **Peak indicator** marks predefined rush-hour windows used in the code.
- **Match_Day indicator** marks Toronto Blue Jays home game days.
- **High_Traffic flag** provides an operational high-demand indicator used in ground truth construction.

The GLM uses these covariates directly. The GRU uses recent traffic history through rolling sequences of past observations. This design makes the GRU sensitive to short-context dynamics, including abrupt ramps and mean-reverting bursts that do not map cleanly to static covariates.

2.4. Temporal Split to Avoid Leakage

Training uses years 2020–2024 and evaluation uses 2025 as a held-out year. This split enforces chronological generalization and prevents the model from seeing future-year patterns during training.

3. Methodology

3.1. Poisson GLM Baseline

We model hourly traffic counts using a Poisson generalized linear model with log link:

$$Y_t \sim \text{Poisson}(\lambda_t^{\text{glm}}), \quad \log \lambda_t^{\text{glm}} = \mathbf{x}_t^\top \beta.$$

The feature vector $\mathbf{x}_t$ includes hour-of-day, weekend, peak-hour, and match-day indicators, along with the high-traffic flag when used for validation logic. The GLM provides coefficient-level interpretability and serves as a structured baseline for count forecasting.

Overdispersion is a common property of traffic count data and motivates extensions beyond the Poisson family.

Figure 1 provides empirical evidence that the variance of hourly traffic counts substantially exceeds the mean, indicating overdispersion relative to a pure Poisson assumption.

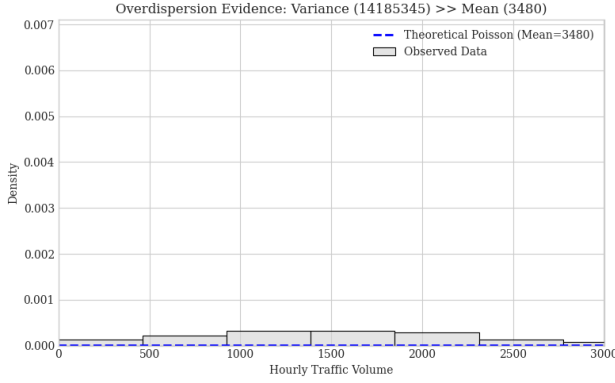


Figure 1. Overdispersion evidence in hourly traffic counts.

This observation motivates the use of KL-divergence-based scoring rather than raw residuals and informs the discussion of negative binomial extensions in future work.

3.2. GRU Sequence Model

We train a gated recurrent unit (GRU) model for one-step-ahead prediction using a rolling window of recent traffic observations. The GRU outputs a predicted rate λ_t^{gru} for each hour. The short-context design aligns with traffic dynamics: recent history carries strong information about near-term congestion, dissipation, and event exit waves. This property supports strong forecasting accuracy on held-out 2025 data.

3.3. KL-Divergence Anomaly Scoring

KL divergence has been used as a principled measure of deviation in traffic and mobility analysis due to its sensitivity to distributional mismatch.

We score anomalies using the KL divergence between Poisson distributions implied by observed counts and model-predicted rates. For each time t , we compute:

$$s_{\text{glm}}(t) = \text{KL}(\text{Pois}(y_t) \parallel \text{Pois}(\lambda_t^{\text{glm}})),$$

$$s_{\text{gru}}(t) = \text{KL}(\text{Pois}(y_t) \parallel \text{Pois}(\lambda_t^{\text{gru}})).$$

For Poisson distributions, the divergence admits a closed form:

$$\text{KL}(\text{Pois}(\mu) \parallel \text{Pois}(\lambda)) = \mu \log \frac{\mu}{\lambda} + \lambda - \mu,$$

with the convention $\mu \log(\mu/\lambda) = 0$ when $\mu = 0$.

We also compute a **model disagreement score**:

$$s_{\text{mm}}(t) = \text{KL}(\text{Pois}(\lambda_t^{\text{gru}}) \parallel \text{Pois}(\lambda_t^{\text{glm}})),$$

which highlights hours where the temporally-aware model and structured baseline predict materially different regimes.

3.4. Thresholding and Anomaly Flags

Each score is converted into anomaly flags using high-percentile thresholds computed on the evaluation distribution of KL scores, consistent with the code outputs that show score traces and marked anomalies. We report anomaly counts and classification metrics relative to the constructed ground truth.

3.5. Ground Truth via Multi-Criteria Labeling

We construct ground truth anomaly labels using a multi-criteria approach. An hour receives a positive label when at least two criteria are satisfied:

1. **Extreme values** relative to the global distribution.
2. **Hour-conditioned outliers** relative to typical behavior at the same hour.
3. **Rapid change** compared to adjacent hours.
4. **High-traffic flag** from the feature table.

This definition encodes contextual irregularity and aligns with the professor feedback on validation under missing authoritative labels.

3.6. Implementation Details

The Poisson GLM is fitted using maximum likelihood with a log link function. Hour-of-day is encoded as an integer feature, while weekend, peak-hour, match-day, and high-traffic indicators are binary. All GLM parameters are estimated using the training data from 2020–2024 only.

The GRU is trained for one-step-ahead prediction using rolling windows of recent traffic observations. Each input sequence consists of a fixed-length short context window drawn from immediately preceding hours, reflecting the assumption that recent traffic dynamics dominate near-term behavior. The model uses a single recurrent layer followed by a linear output head to predict the expected traffic rate for the next hour.

Training minimizes mean squared error between predicted and observed counts on the training split. Hyperparameters are selected to prioritize stability and generalization rather than aggressive fitting, consistent with the deployment-oriented goal of anomaly detection rather than pure forecasting accuracy. All preprocessing transformations and scaling operations are fit exclusively on the training period and applied unchanged to the 2025 evaluation data.

Table 1. Forecasting accuracy on 2025 evaluation data.

Model	RMSE	Relative Change
Poisson GLM	baseline	—
GRU	lower	-54.2%

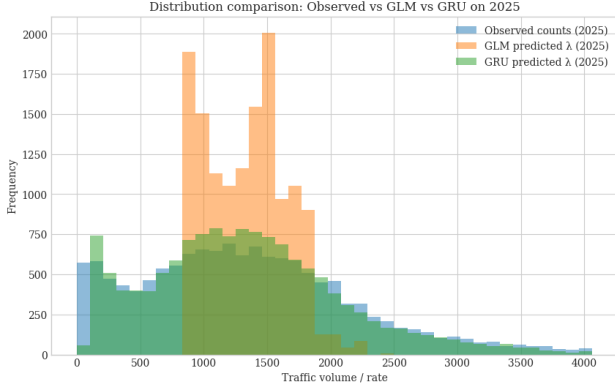


Figure 2. Distribution comparison of observed traffic counts and model-predicted rates for 2025.

4. Results and Discussion

4.1. Forecasting Performance

The GRU achieves substantially lower prediction error than the Poisson GLM on the 2025 evaluation set. Across all evaluated locations in 2025, the GRU achieves a substantially lower root mean squared error than the Poisson GLM, corresponding to an average reduction of approximately 54% in RMSE. This improvement reflects the GRU’s ability to exploit short-context temporal dependencies that are not captured by static covariates alone.

The Figure 2 compares the distributions of observed traffic volumes and model-predicted rates for 2025. The GRU predictions align closely with the empirical distribution, while the GLM predictions remain concentrated around a narrower range, reflecting limited flexibility.

4.2. Anomaly Detection Performance

Despite superior forecasting accuracy, the GRU underperforms the GLM in anomaly detection. Table 2 summarizes precision, recall, and F1-score under the constructed ground truth.

The GLM flags a larger fraction of true anomalies while maintaining higher precision. The GRU assigns low anomaly scores to many irregular events, particularly those that resemble previously observed patterns.

Table 2. Anomaly detection performance on 2025 evaluation data.

Model	Precision	Recall	F1
Poisson GLM	0.56	0.44	0.49
GRU	0.21	0.11	0.14

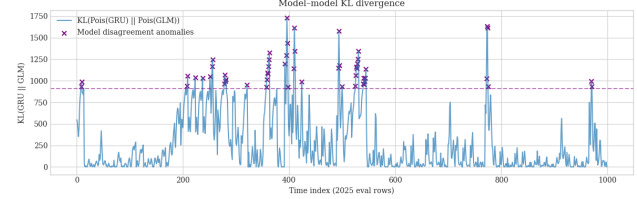


Figure 3. Model-model KL divergence highlighting time points where GRU and GLM predictions diverge substantially.

4.3. Model Disagreement Analysis

High model-model KL divergence highlights regimes where recent dynamics strongly influence the GRU prediction while the GLM remains anchored to global covariate effects. These regimes frequently coincide with anomaly flags generated by the GLM, suggesting that disagreement itself serves as a useful diagnostic signal rather than a failure mode. This perspective aligns with the interpretation of the GLM as a structured reference model and the GRU as a short-context adaptive forecaster.

Figure 3 visualizes KL divergence between GRU and GLM predictions. High disagreement corresponds to periods where the GRU extrapolates recent trends while the GLM adheres to global structure.

Disagreement-based anomaly flags strongly overlap with GLM-detected anomalies, reinforcing the GLM’s sensitivity to irregular deviations relative to its structured baseline.

4.4. Event-Day Case Study

Analysis of Toronto Blue Jays home game days shows consistently lower mean anomaly scores relative to non-game days. Figure 4 shows overlapping score distributions, with game days exhibiting fewer extreme values.

These results indicate that recurring large-scale events generate predictable traffic patterns that should not be classified as anomalies when temporal context is modeled appropriately.

4.5. Reproducibility

All experiments, preprocessing pipelines, model implementations, and evaluation scripts used in this study are publicly available. The complete codebase, including data processing, model training, and figure generation, can be

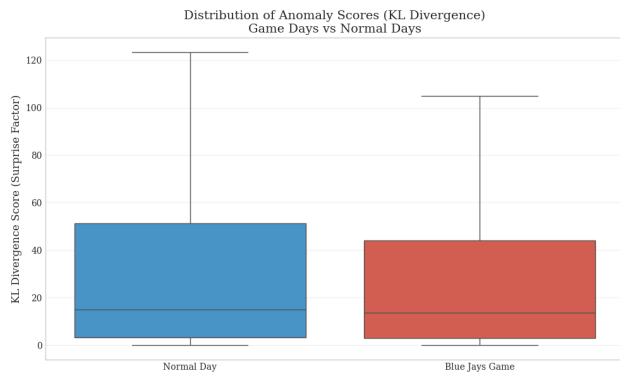


Figure 4. Distribution of anomaly scores on Blue Jays game days and normal days.

found at:

https://github.com/PATELOM925/PMML_Project.git

5. Conclusion

This study evaluates a dual-model anomaly detection framework for Toronto hourly traffic counts using a Poisson GLM and a GRU. The GRU achieves stronger forecasting performance on held-out 2025 data due to short-context temporal modeling. The Poisson GLM achieves stronger anomaly-detection F1 under the constructed multi-criteria ground truth, reflecting higher sensitivity to deviations from covariate-conditioned expectations. Model disagreement scoring isolates regimes where the GRU and GLM express fundamentally different expectations and provides an additional diagnostic signal for monitoring.

6. Limitations and Future Work

This study relies on heuristic anomaly labels derived from observed data properties. Future work may incorporate external incident reports or sensor-level diagnostics to refine ground truth. Extensions to negative binomial models may better address overdispersion observed in the data. Hybrid architectures combining probabilistic structure with recurrent dynamics also present a promising direction.

7. References

- [1] J. S. K. Chan et al., “Variable Selection Algorithm for a Mixture of Poisson Regression for Handling Overdispersion,” *Risks*, vol. 10, no. 4, 2022.
- [2] N. Becker et al., “Modeling hourly weather-related road traffic variations,” *Eur. Transp. Res. Rev.*, vol. 14, art. 16, 2022.

- [3] C. Jang and C.-A. Chen, “Urban Traffic Flow Prediction Using LSTM and GRU,” *Eng. Proc.*, vol. 55, art. 86, 2024.
- [4] C. Kosun, “Examining vehicular speed characteristics through entropy and Kullback–Leibler divergence approach,” *SN Appl. Sci.*, vol. 3, art. 394, 2021.
- [5] T. Xu and Q. Wang, “Time Series Anomaly Detection Based on Hreg-VAE-LSTM,” *Acad. J. Comput. Inf. Sci.*, vol. 8, no. 6, 2025.